



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Setup your Local Development IDE

You need the following set of requirements for Spark setup on your machine. Let us go through this setup one by one.



Requirements for Spark setup on Windows

- JDK 8 or JDK 11
- Python 3.6 or higher
- Hadoop WinUtils
- Spark Binaries
- Environment Variables
- Python IDE

Let's start with the JDK installation.

Spark is a JVM application. So you need JDK on your local machine for installing and running your Spark application.

The current most recent Java is at JDK 19. However, as of today, Spark supports JDK 8 or JDK 11.

Do not use any other version of JDK because Spark is not well tested on other versions.

You might face problems running Spark on other JDK versions.



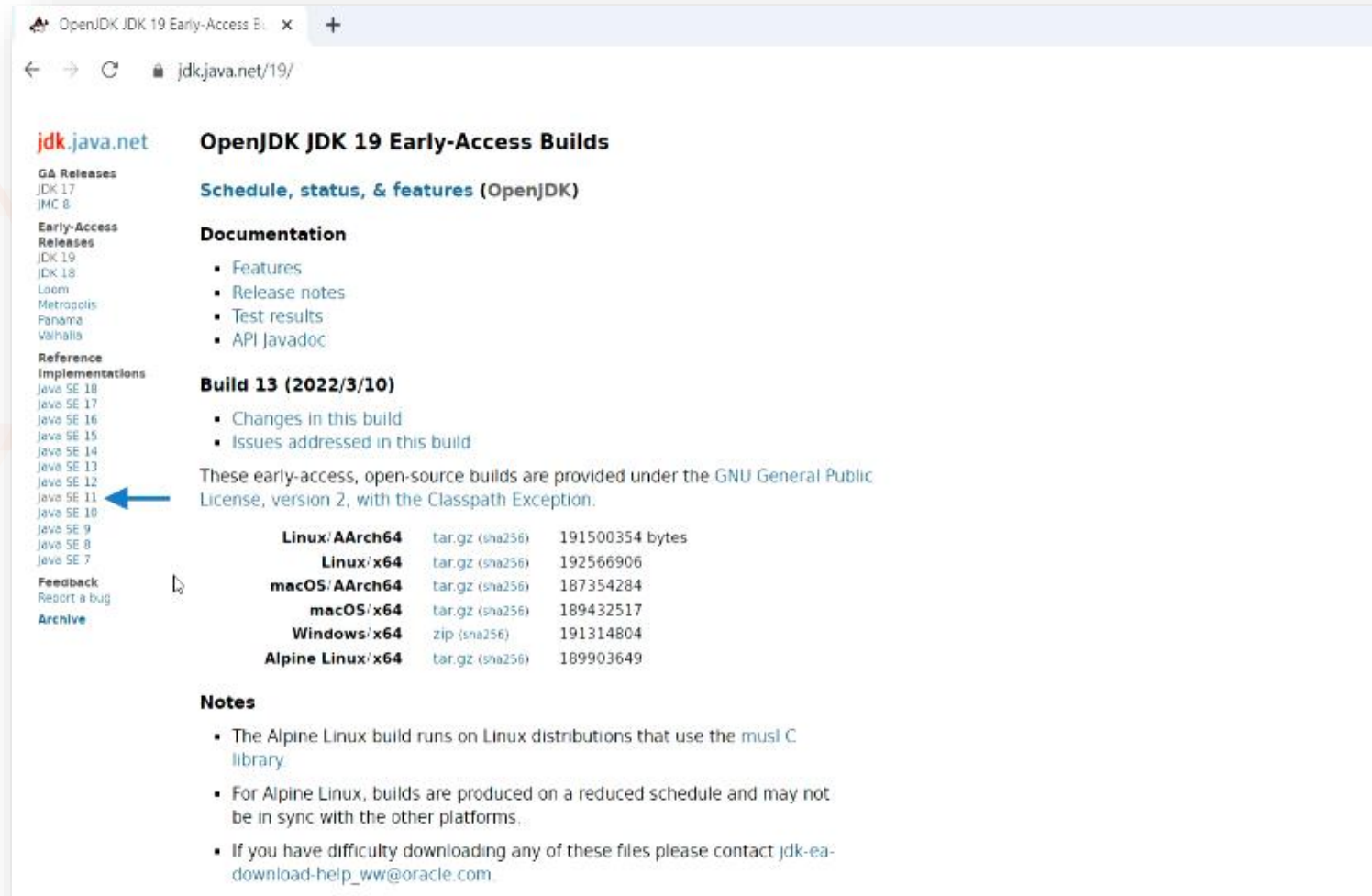
Requirements for Spark setup on Windows

- • JDK 8 or JDK 11
- Python 3.6 or higher
- Hadoop WinUtils
- Spark Binaries
- Environment Variables
- Python IDE

Start your browser and visit <http://jdk.java.net/>
You will a link for the most recent version of the JDK. Click the link to go to the downloads page.



Then choose Java SE 11 from the list of all available versions.
And download the JDK 11 for your platform (Windows/Linux) in the next step.



jdk.java.net

OpenJDK JDK 19 Early-Access Builds

[Schedule, status, & features \(OpenJDK\)](#)

Documentation

- [Features](#)
- [Release notes](#)
- [Test results](#)
- [API Javadoc](#)

Build 13 (2022/3/10)

- [Changes in this build](#)
- [Issues addressed in this build](#)

These early-access, open-source builds are provided under the GNU General Public License, version 2, with the Classpath Exception.

Linux/AArch64	tar.gz (sha256)	191500354 bytes
Linux/x64	tar.gz (sha256)	192566906
macOS/AArch64	tar.gz (sha256)	187354284
macOS/x64	tar.gz (sha256)	189432517
Windows/x64	zip (sha256)	191314804
Alpine Linux/x64	tar.gz (sha256)	189903649

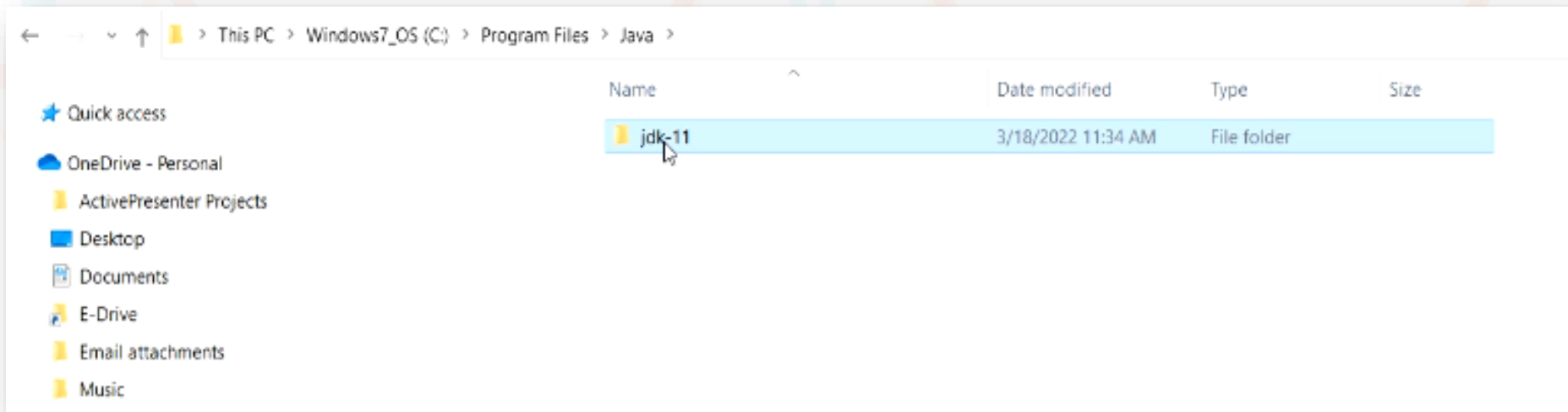
Notes

- The Alpine Linux build runs on Linux distributions that use the [musl C library](#).
- For Alpine Linux, builds are produced on a reduced schedule and may not be in sync with the other platforms.
- If you have difficulty downloading any of these files please contact jdk-ea-download-help_ww@oracle.com.

Go to the downloaded file and extract it. You will see the JDK-11 folder.

Inside the JDK-11, you will see some more folders. Copy the JDK-11 folder and paste it at some permanent place.

It is recommended to keep it in *C:\Program Files\Java* folder.



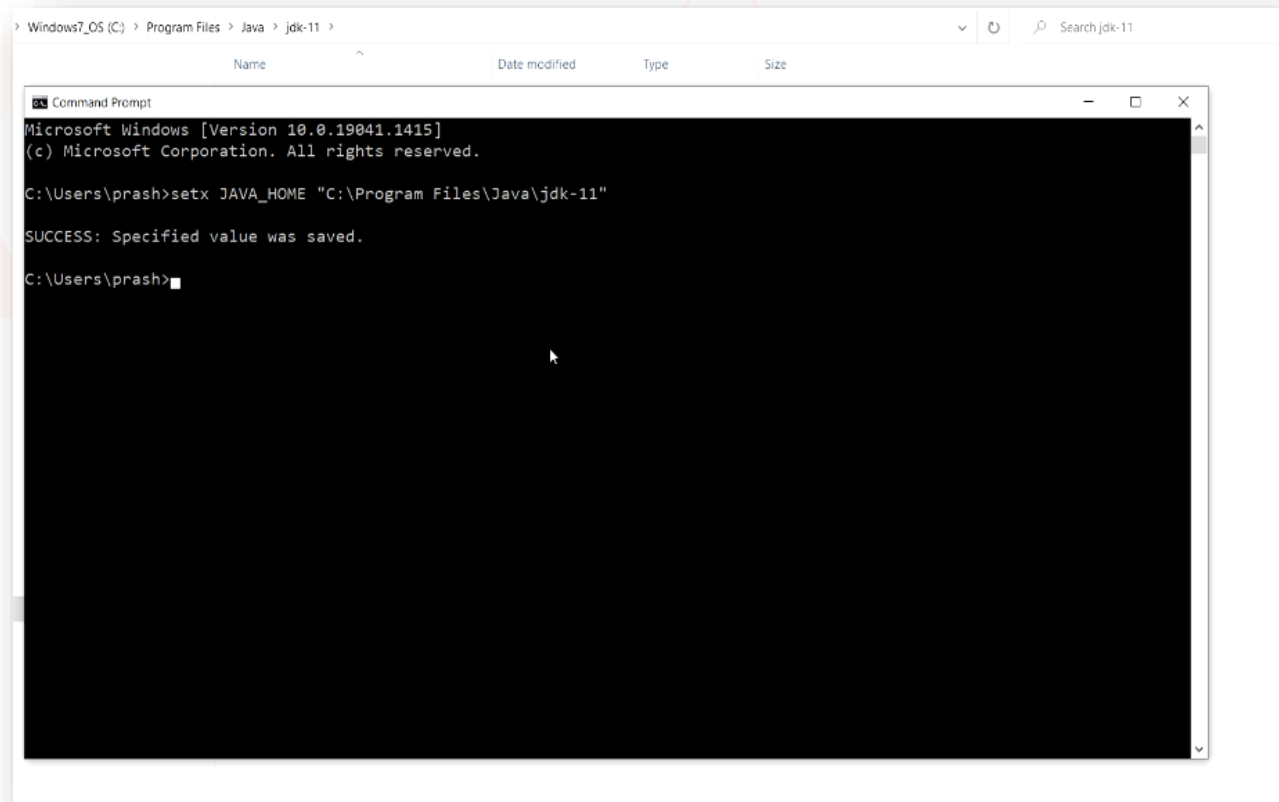
You have JDK 11 on your system. But it won't work as we need to set up two environment variables to make it work.

Start windows command prompt.

Now you can use the setx command as shown below to set the JAVA_HOME environment variable.

The JAVA_HOME variable must point to your JDK-11 directory.

Make sure you see a success message.



The image shows a Windows File Explorer window in the background with the address bar set to 'C:\Program Files\Java\jdk-11'. In the foreground, a Command Prompt window is open, displaying the following text:

```
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

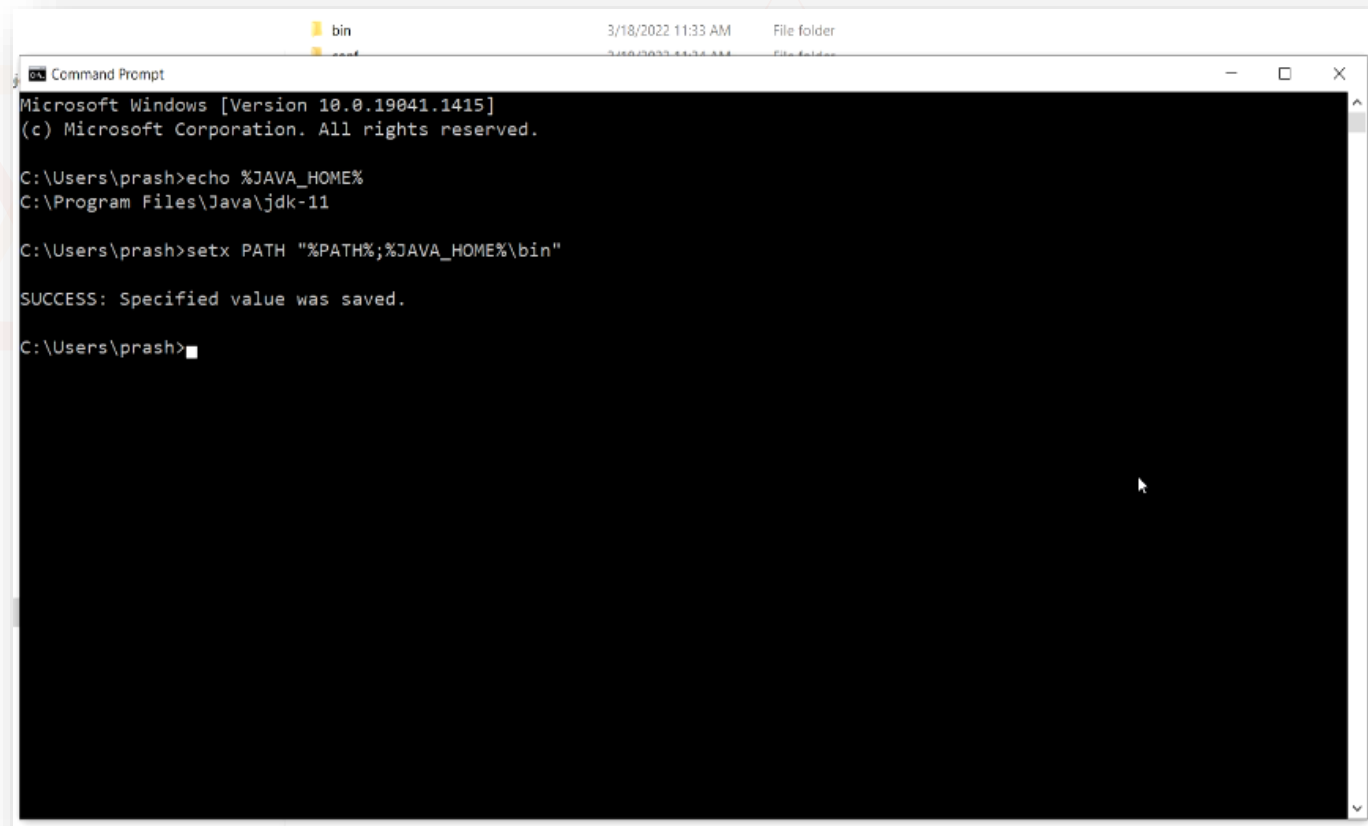
C:\Users\prash>setx JAVA_HOME "C:\Program Files\Java\jdk-11"

SUCCESS: Specified value was saved.

C:\Users\prash>
```


If you want to check that your environment variable is set properly, you can restart the command prompt and check it using the echo command as shown below.

Then the second requirement is to add the JAVA_HOME\bin to your PATH environment variable. You can use the setx command as shown in the image below. And make sure you get a success message. Ensure you include the current value of your PATH environment variable and add JAVA_HOME\bin to the same.



```
bin
3/18/2022 11:33 AM
File folder
File folder

Command Prompt
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

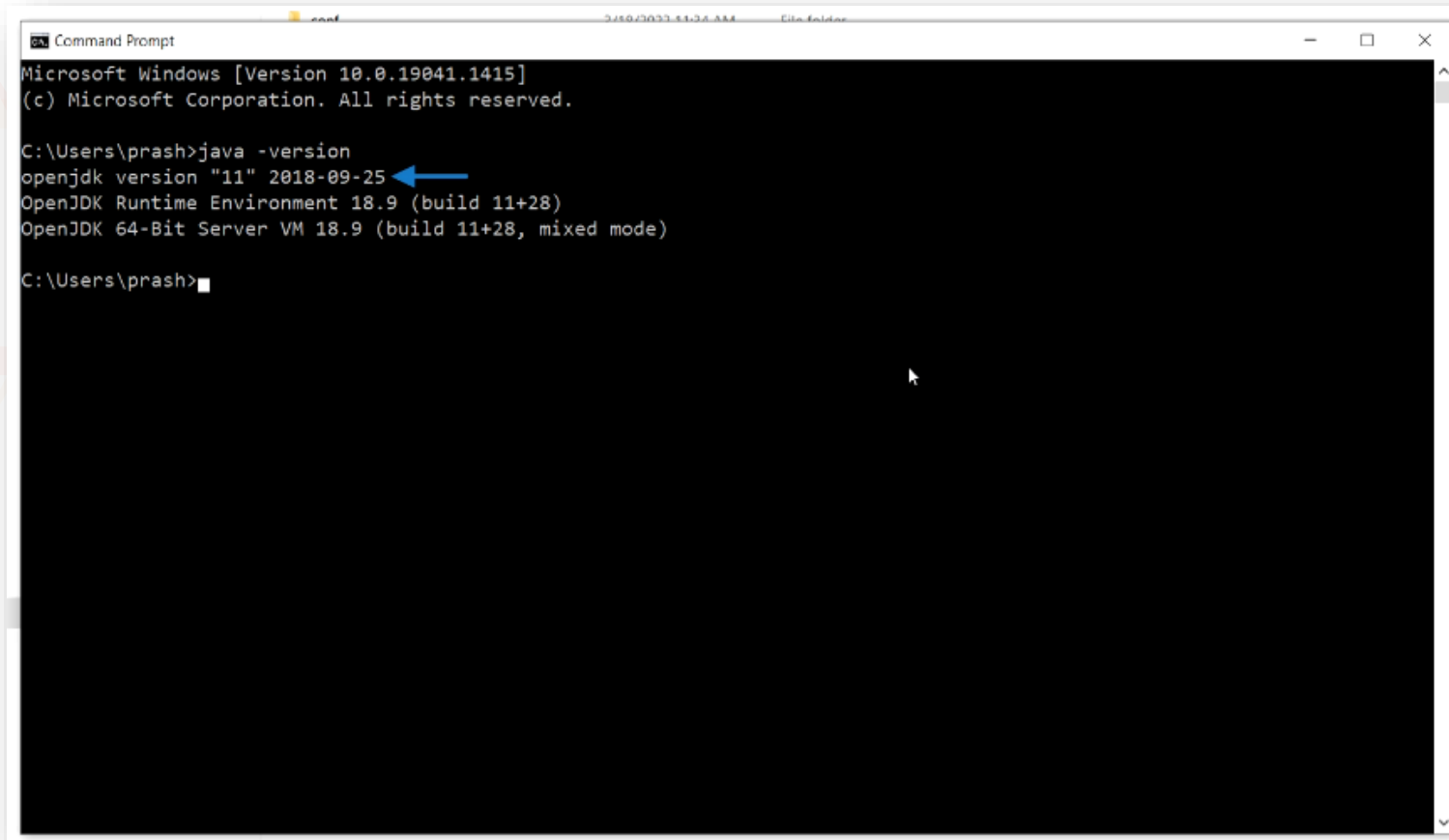
C:\Users\prash>echo %JAVA_HOME%
C:\Program Files\Java\jdk-11

C:\Users\prash>setx PATH "%PATH%;%JAVA_HOME%\bin"

SUCCESS: Specified value was saved.

C:\Users\prash>
```

Finally, execute the Java -version command, and you should see the current Java version. Make sure you see Java version 11.
If you already have a Java 8 or Java 11 pre-installed on your machine, you can skip the Java installation steps.



```
Command Prompt
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prash>java -version
openjdk version "11" 2018-09-25
OpenJDK Runtime Environment 18.9 (build 11+28)
OpenJDK 64-Bit Server VM 18.9 (build 11+28, mixed mode)

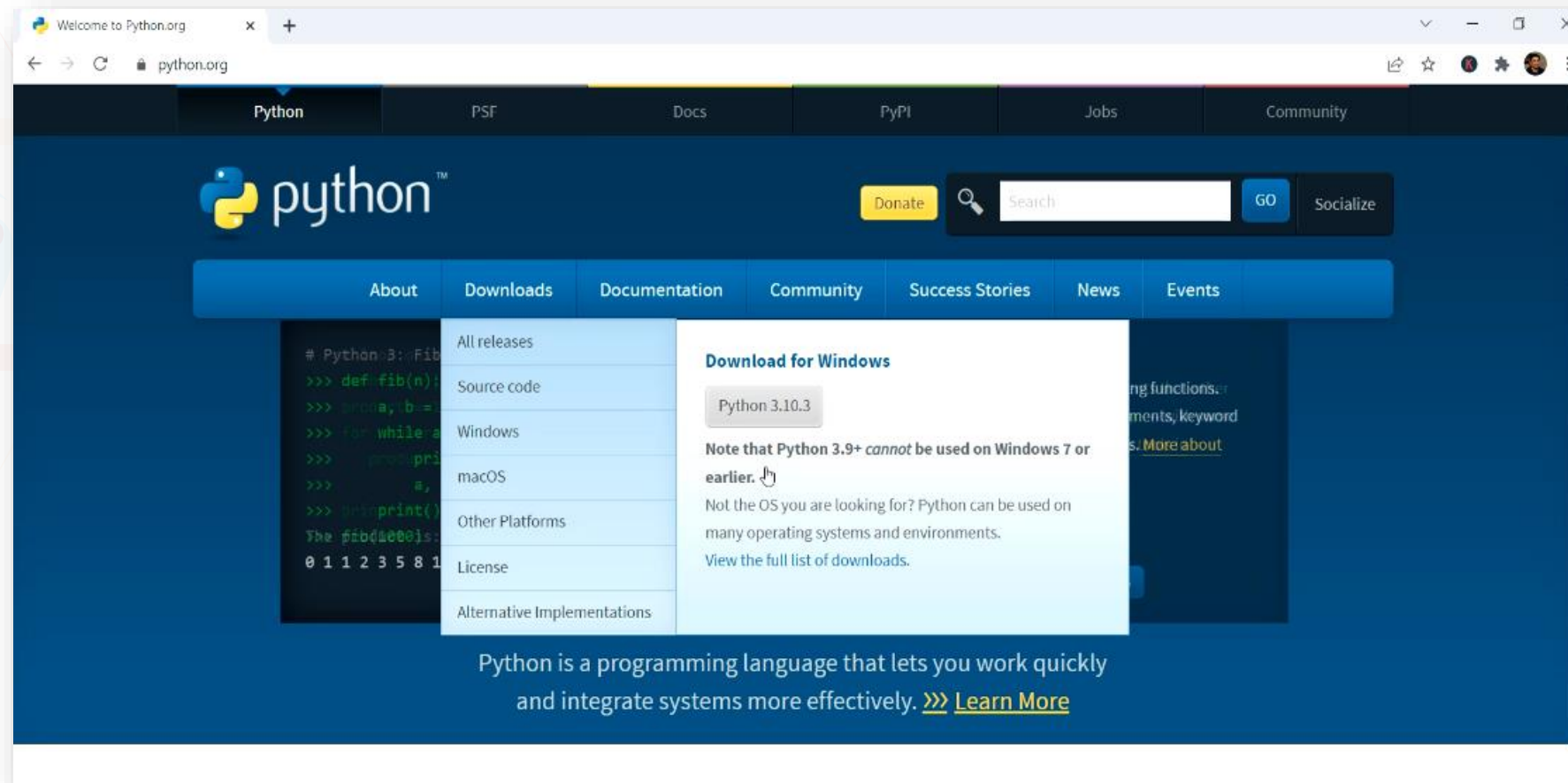
C:\Users\prash>
```

So, basically there are three steps you need to check in order to complete your JDK installation.

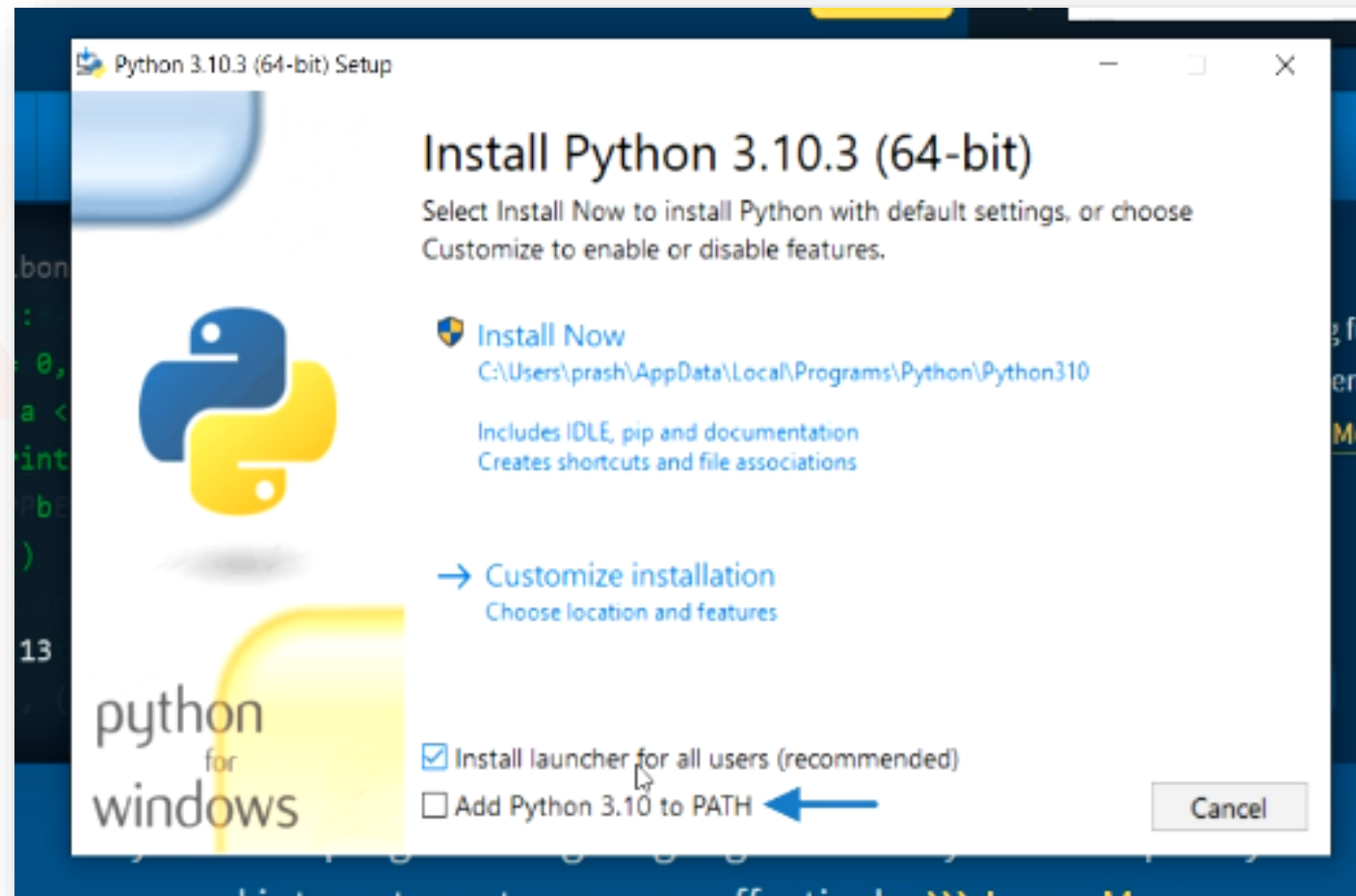
1. JAVA_HOME environment variable is set, and it is pointing to Java 8 or java 11
2. JAVA_HOME\bin is included in your PATH environment variable.
3. java -version command is showing Java 8 or Java 11

If all these three are there in place, you are done.

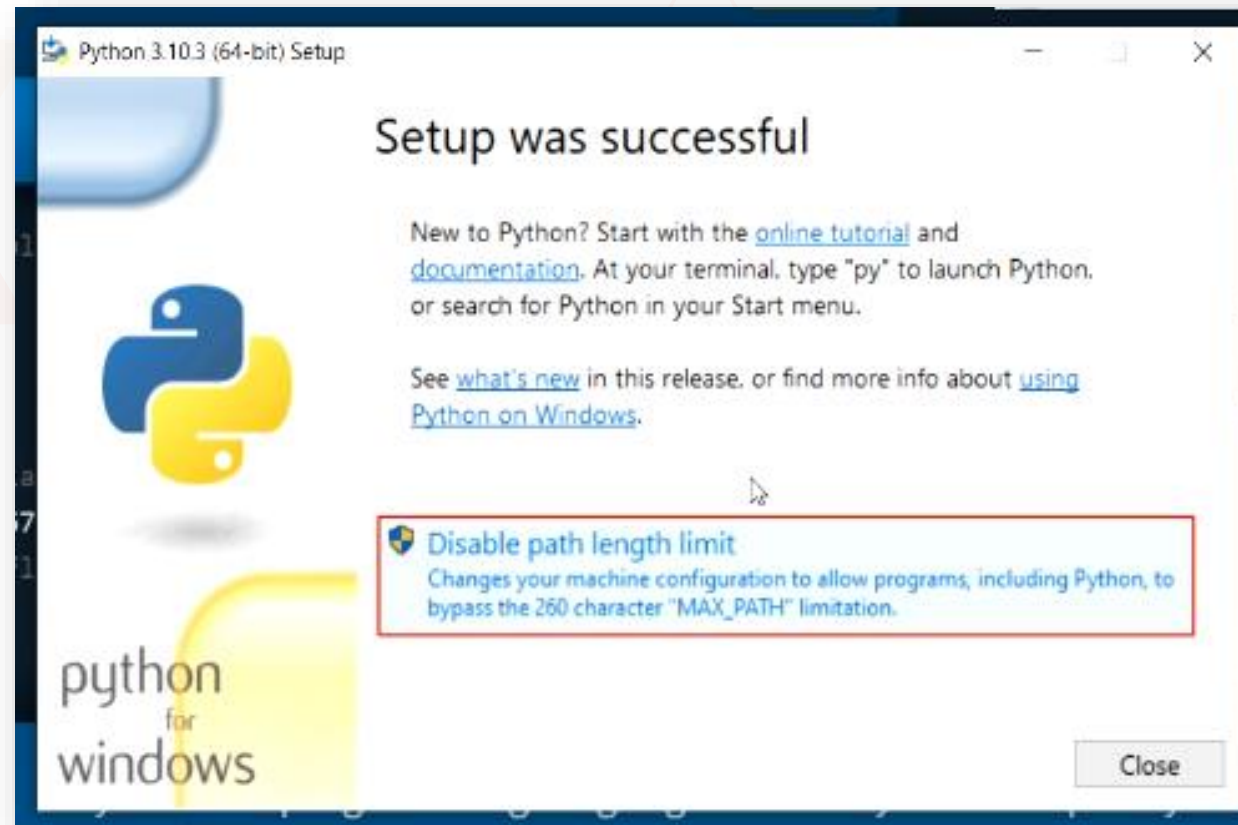
The next step to setup your Local Development IDE is to install Python. Visit <https://www.python.org/> and download the latest Python version for your machine.



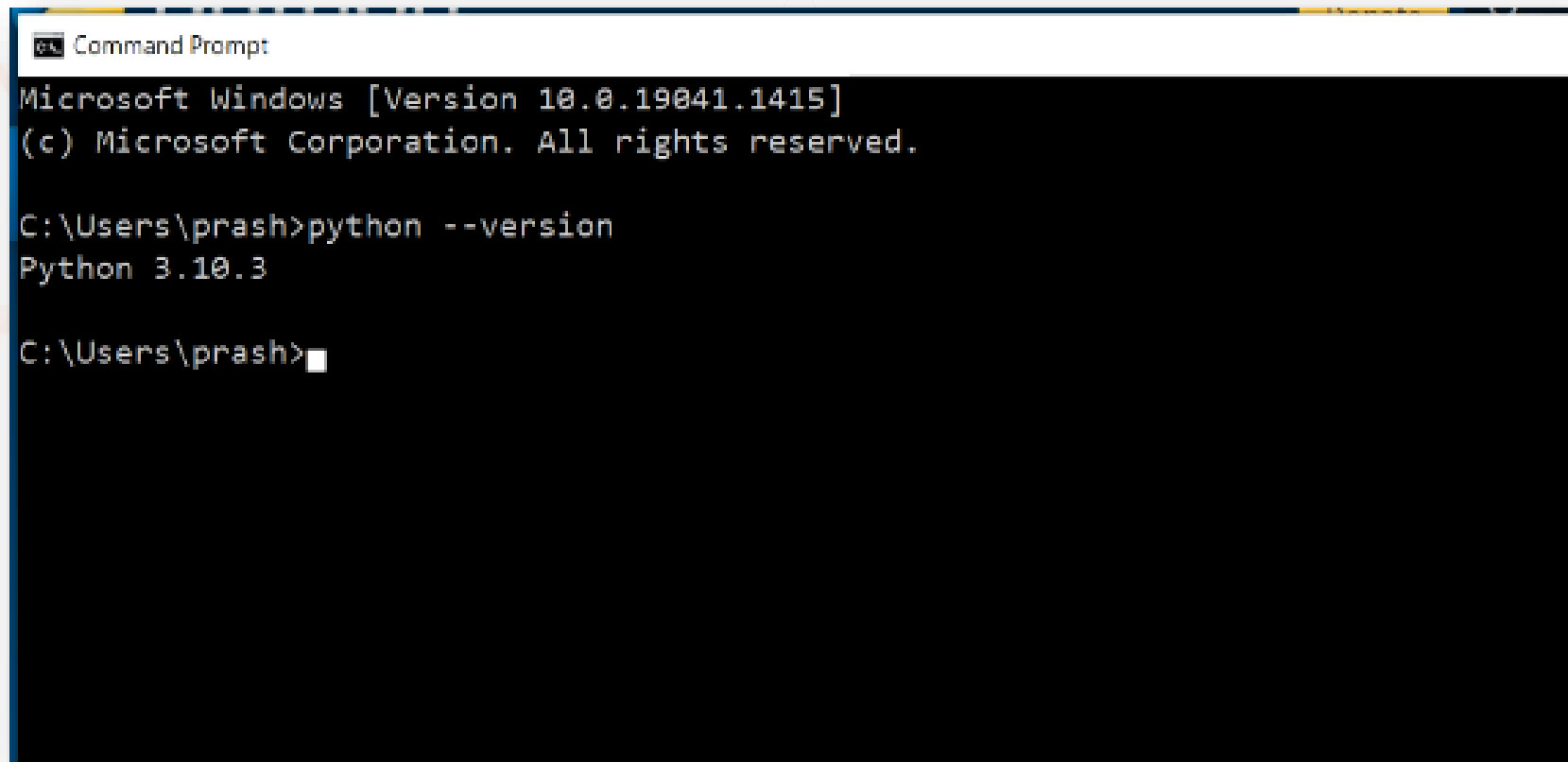
Once downloaded, click the file to run it. You should see an option to add Python to your PATH environment variable. Enable that option, so we do not have to do it manually. Then click the install now button.



It might take a couple of minutes to complete the installation. You might see a notification to disable the path length limit. An older windows machine allows only 255 characters for any path. The max path length limit is a problem, and this limitation is removed from the newer systems. If you are using an older system, you might see this message. I recommend that you choose this option and remove the max path length.



Once done with the installation, you should check it once. Start a command prompt and run *python --version* command and make sure you see the latest version that you recently installed.

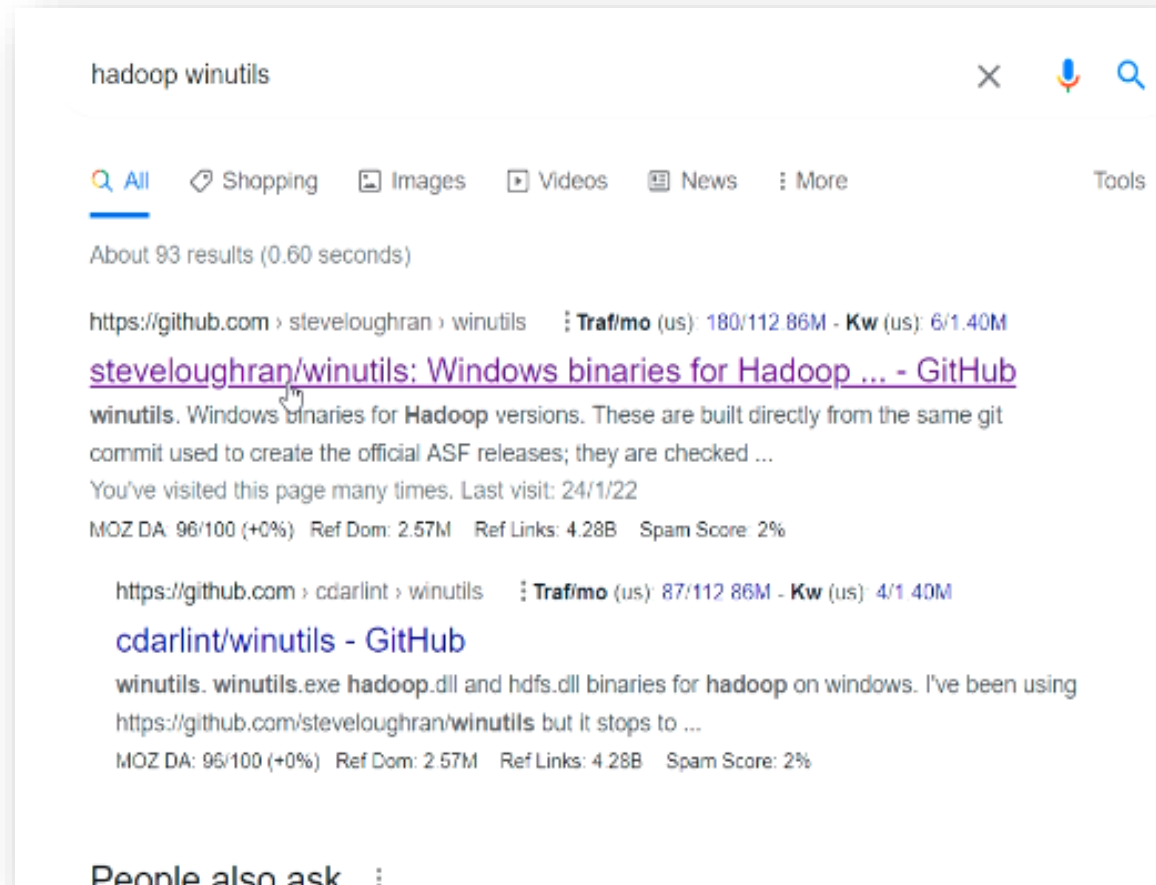
A screenshot of a Windows Command Prompt window. The title bar reads 'Command Prompt'. The window content shows the following text: 'Microsoft Windows [Version 10.0.19041.1415]', '(c) Microsoft Corporation. All rights reserved.', 'C:\Users\prash>python --version', 'Python 3.10.3', and 'C:\Users\prash>'. The prompt is at the end of the last line, indicated by a small white square.

```
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

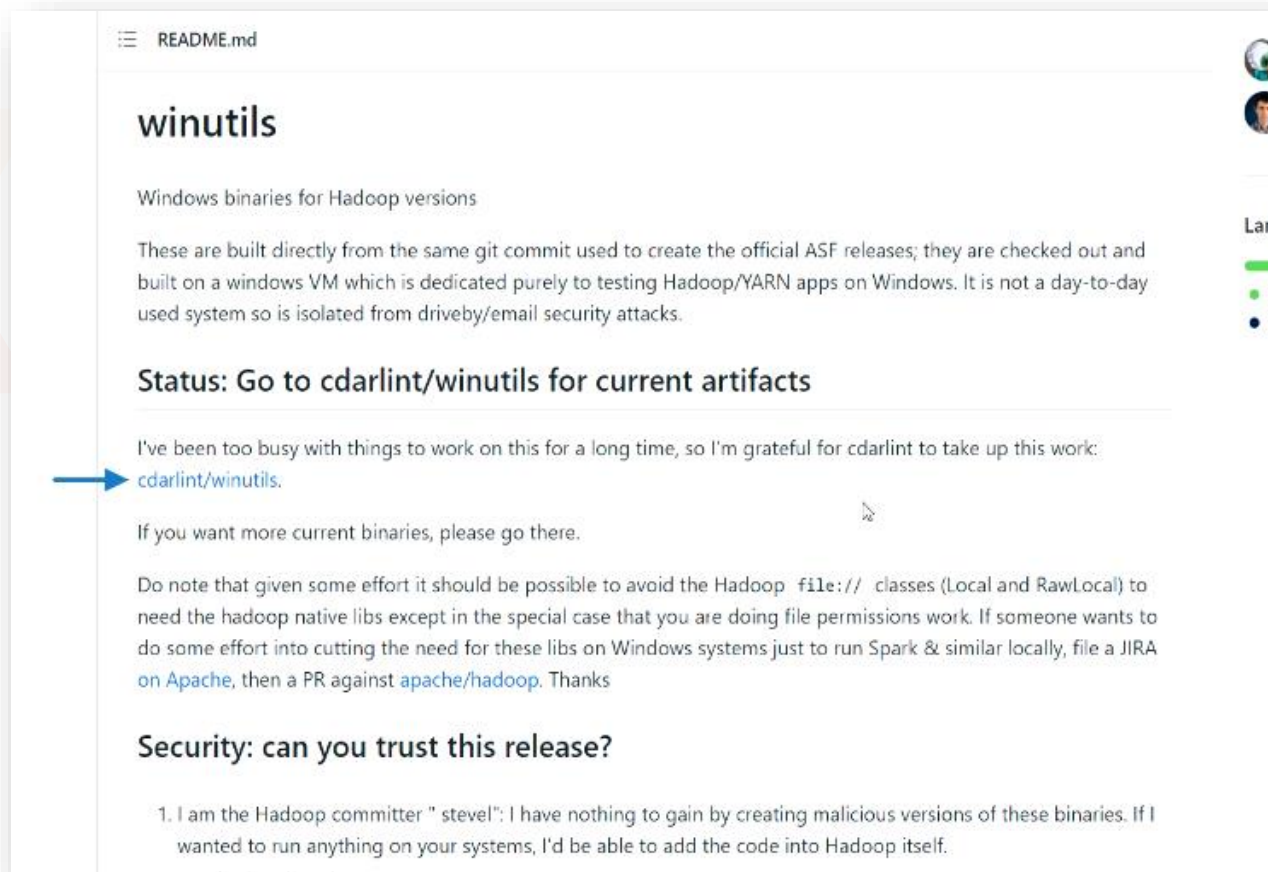
C:\Users\prash>python --version
Python 3.10.3

C:\Users\prash>
```


The following requirement is to setup Hadoop Winutils. Google for Hadoop Winutils, and you should see a GitHub page link. This is an old repository and is not being actively managed.

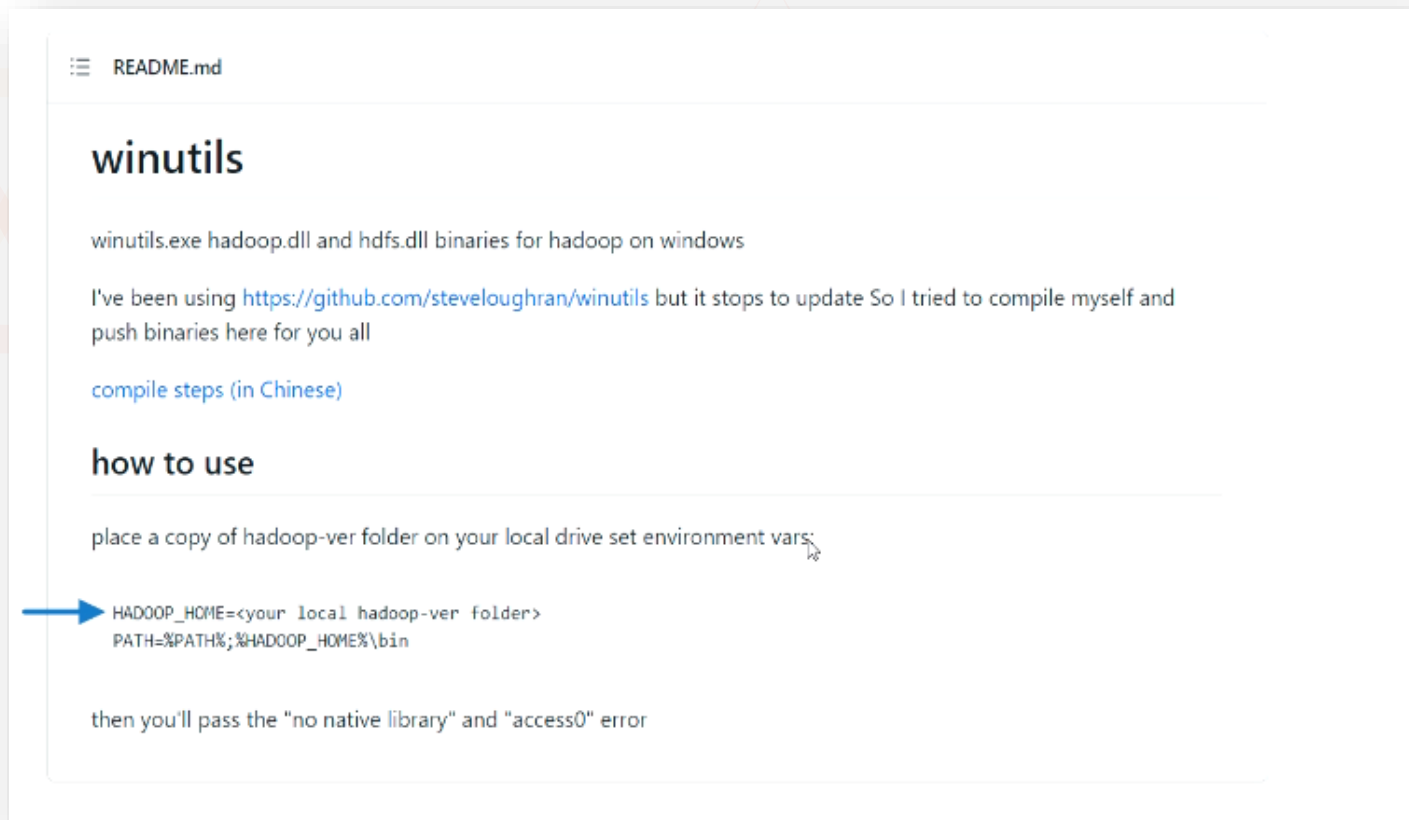


After following the GitHub link, scroll down, and you will see a link for the new repository as shown in the image below. Follow the link to go to the current repository.

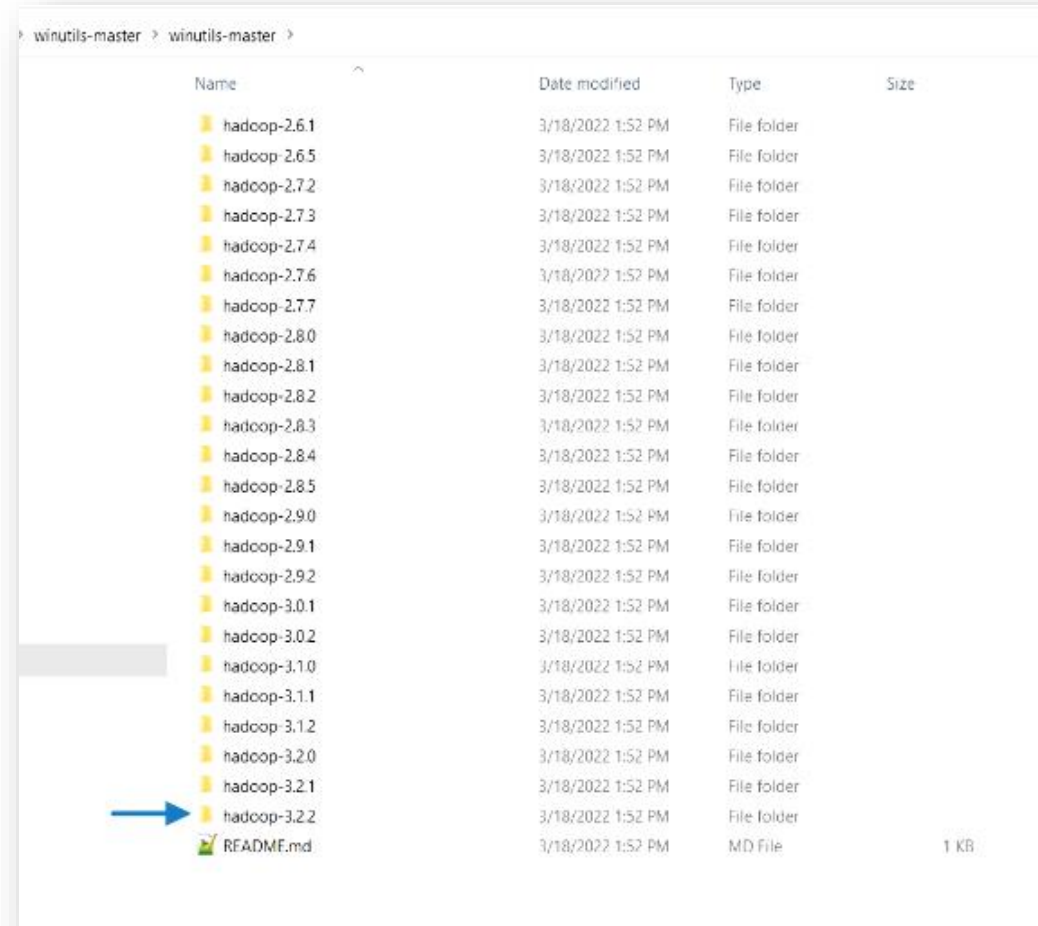


Download the current repository, and if you scroll down you will see 2 steps of using this:

1. Setup your HADOOP_HOME environment variable
2. Include HADOOP_HOME\bin to your PATH environment variable.

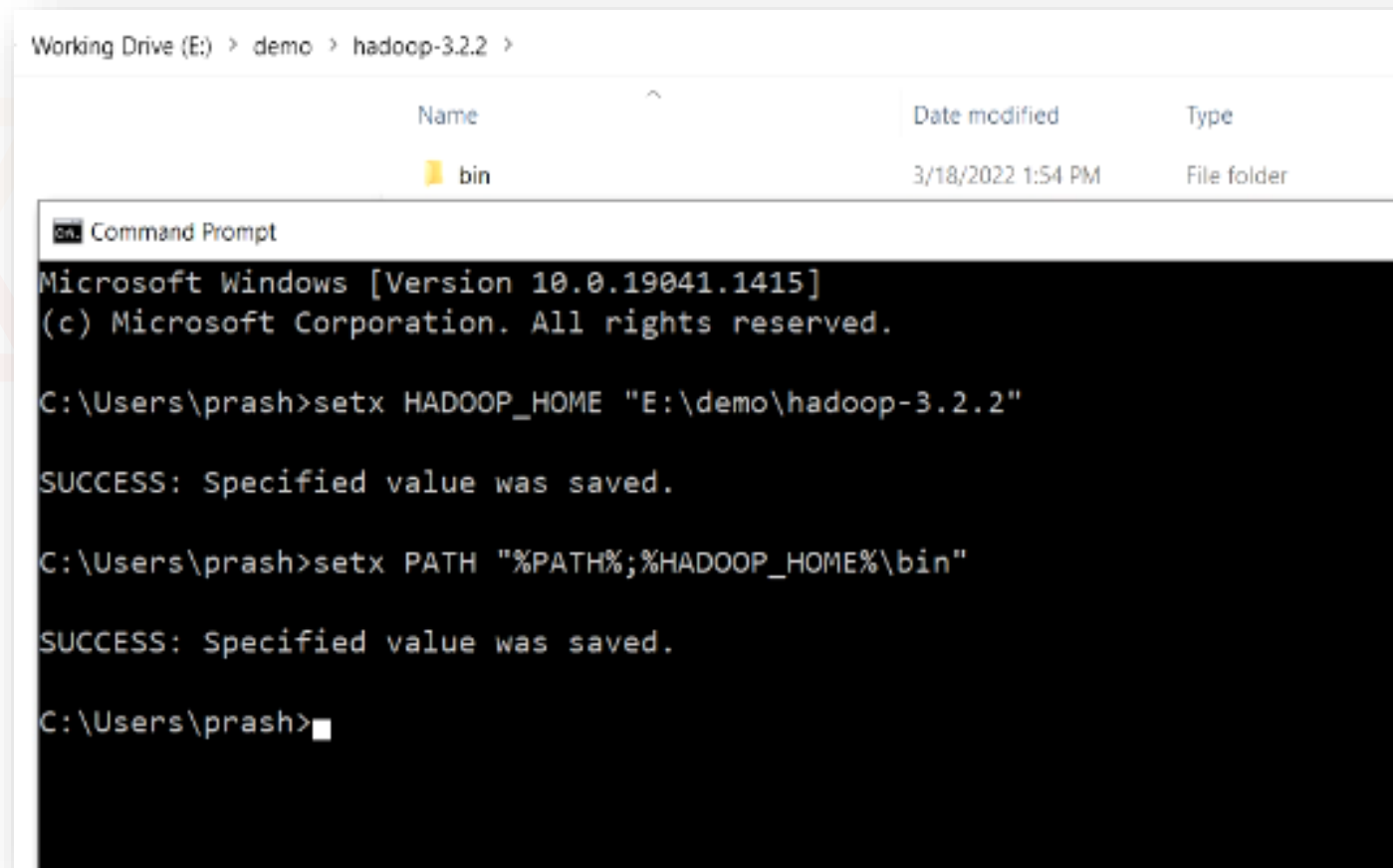


Once downloaded, go to the zip file and un-compress it. You will see the winutils-master directory. Go inside, and you will see many other folders. So I recommend copying the latest version folder and pasting it at another permanent location. If you go inside, and you will see a bin directory.



The next step is to setup your environment variable. Go the command prompt and use the setx command to set the environment variable, as shown in the image below. Make sure you see the success message.

Now add it to the PATH environment variable, as shown in the image below. Make sure you see the success message again.



The screenshot shows a Windows File Explorer window with the address bar set to 'Working Drive (E:) > demo > hadoop-3.2.2'. The main pane displays a table with columns 'Name', 'Date modified', and 'Type'. A single entry is shown: 'bin' (File folder) with a date modified of '3/18/2022 1:54 PM'. Overlaid on this is a Command Prompt window titled '04. Command Prompt'. The Command Prompt shows the following text:

```
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prash>setx HADOOP_HOME "E:\demo\hadoop-3.2.2"

SUCCESS: Specified value was saved.

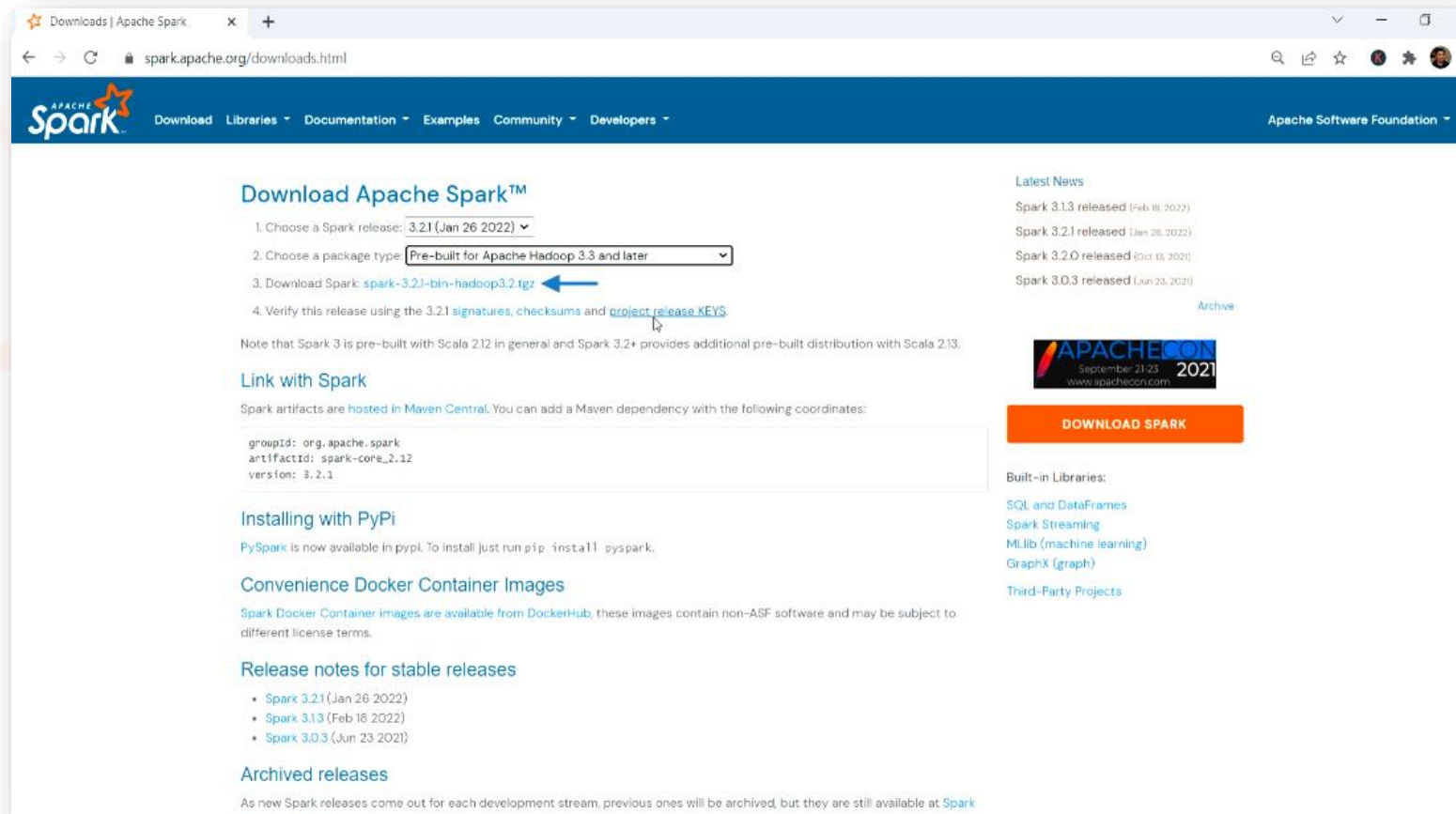
C:\Users\prash>setx PATH "%PATH%;%HADOOP_HOME%\bin"

SUCCESS: Specified value was saved.

C:\Users\prash>
```

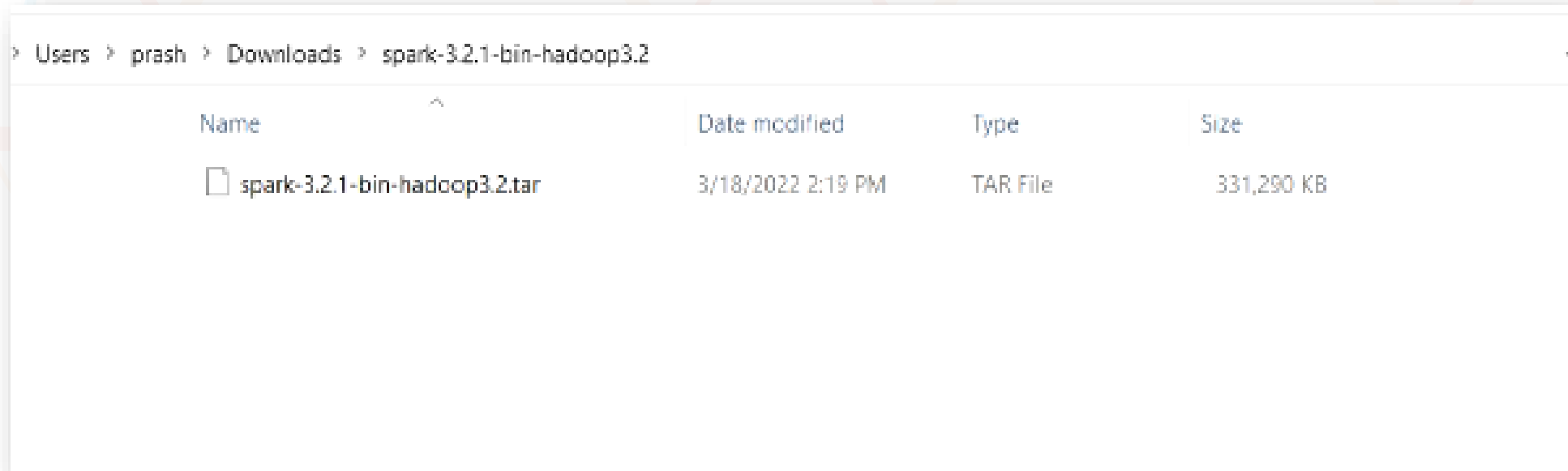
The next requirement is Spark Binaries.

Visit <https://spark.apache.org/> and click the download link. Then choose the latest Spark version and the Hadoop version. Now you can click the download link and start the download.



Once downloaded, go to the file and uncompress it. The Spark binary is not a zip file. It is a tgz file.

So you might need 7zip for uncompressing this tgz on the windows platform. After compressing the file, go inside, and you will see a tar file. This one is also a compressed file. You can untar this file using the 7zip tool.

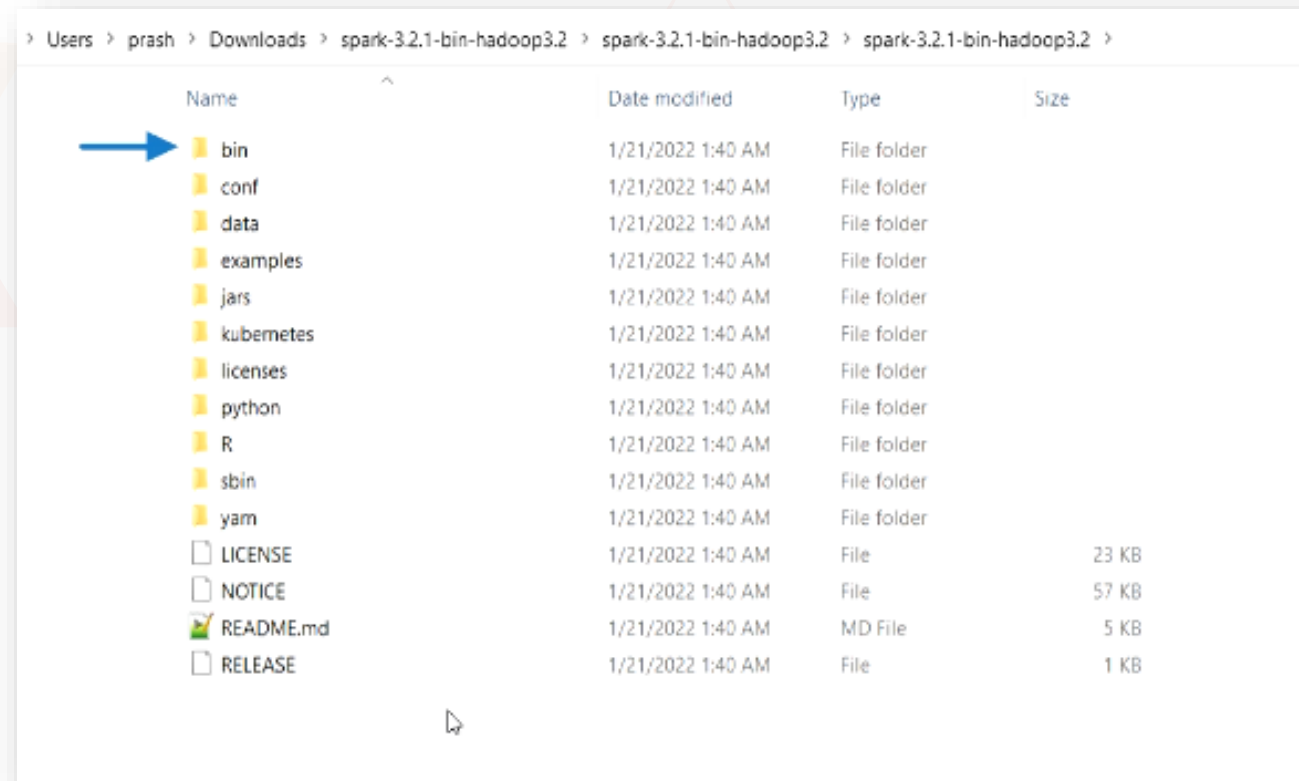


The screenshot shows a file explorer window with the following path: > Users > prash > Downloads > spark-3.2.1-bin-hadoop3.2. The window displays a table of files and folders.

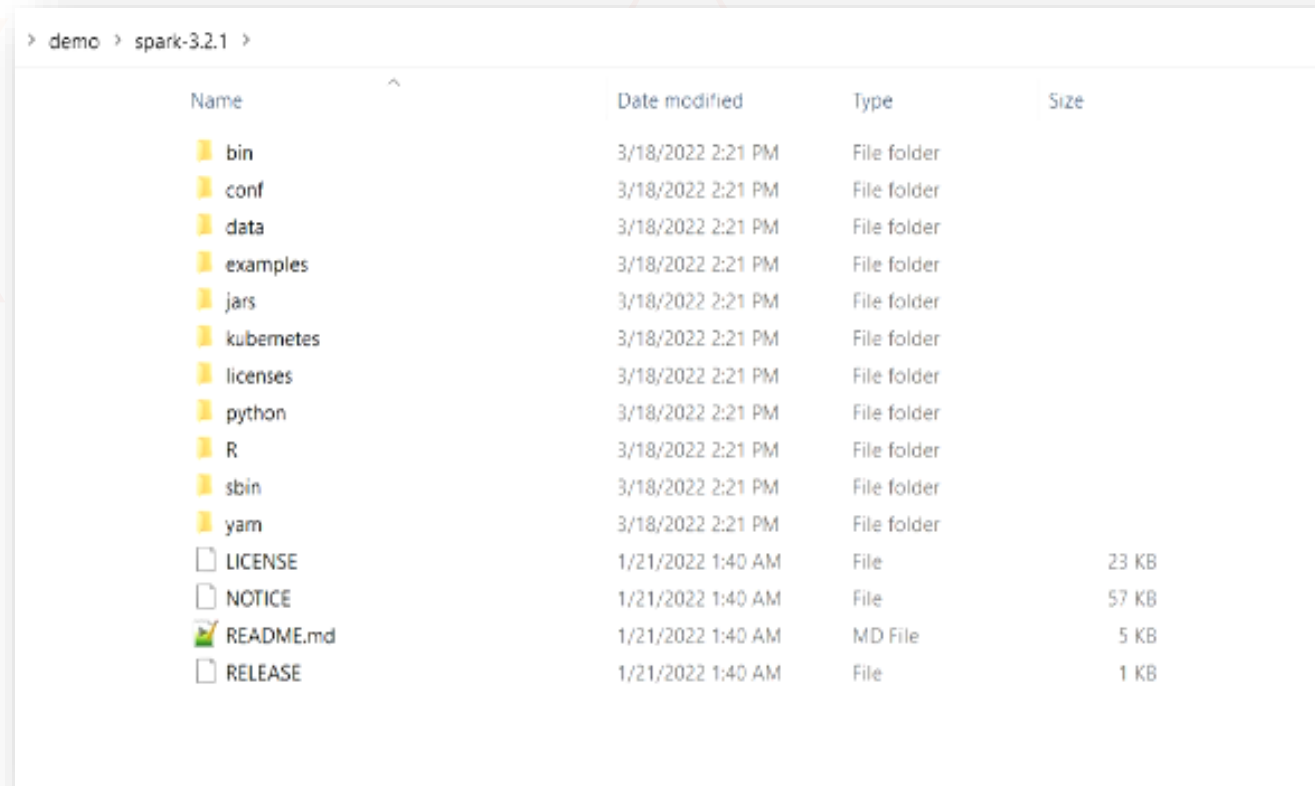
Name	Date modified	Type	Size
 spark-3.2.1-bin-hadoop3.2.tar	3/18/2022 2:19 PM	TAR File	331,290 KB

After compressing the tar file, you will see another uncompressed directory.

If you go inside the folder a couple of times you will see the bin directory shown in the image below.

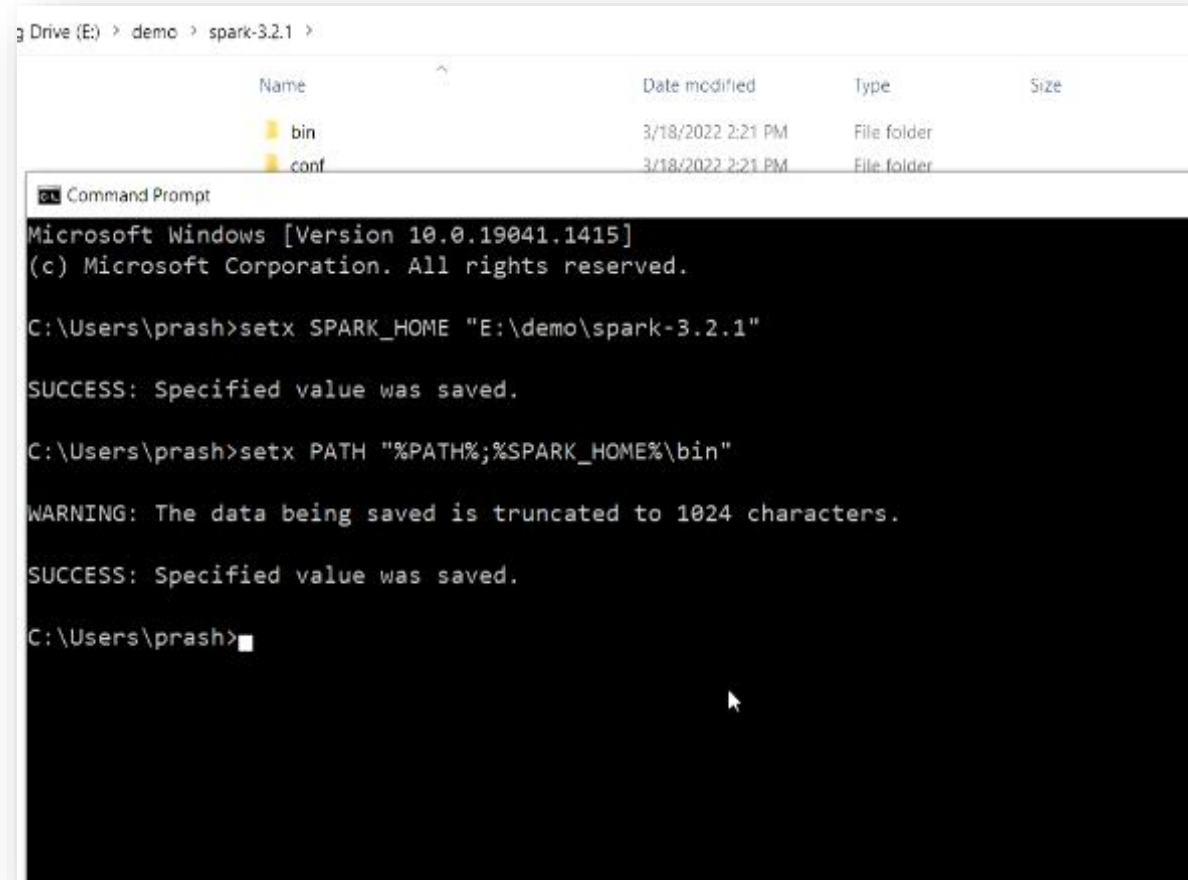


There are too much folders inside folders, so you can copy the parent folder and paste it to a permanent location. So the image shown below is your spark home. But we must set the environment variables.



> demo > spark-3.2.1 >				
Name	Date modified	Type	Size	
bin	3/18/2022 2:21 PM	File folder		
conf	3/18/2022 2:21 PM	File folder		
data	3/18/2022 2:21 PM	File folder		
examples	3/18/2022 2:21 PM	File folder		
jars	3/18/2022 2:21 PM	File folder		
kubernetes	3/18/2022 2:21 PM	File folder		
licenses	3/18/2022 2:21 PM	File folder		
python	3/18/2022 2:21 PM	File folder		
R	3/18/2022 2:21 PM	File folder		
sbin	3/18/2022 2:21 PM	File folder		
yam	3/18/2022 2:21 PM	File folder		
LICENSE	1/21/2022 1:40 AM	File	23 KB	
NOTICE	1/21/2022 1:40 AM	File	57 KB	
README.md	1/21/2022 1:40 AM	MD File	5 KB	
RELEASE	1/21/2022 1:40 AM	File	1 KB	

Start your command prompt and use the setx command to set the SPARK_HOME environment variable. Make sure you see the success message. Then you should also add the SPARK_HOME\bin to our PATH environment variable. You might also see a warning that data being saved is truncated. If you see it, your PATH environment variable is not set correctly.



The screenshot shows a Windows File Explorer window with the address bar set to 'g Drive (E:) > demo > spark-3.2.1'. The file list contains two folders: 'bin' and 'conf', both modified on 3/18/2022 at 2:21 PM. Overlaid on this is a Windows Command Prompt window. The Command Prompt displays the following text:

```
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prash>setx SPARK_HOME "E:\demo\spark-3.2.1"

SUCCESS: Specified value was saved.

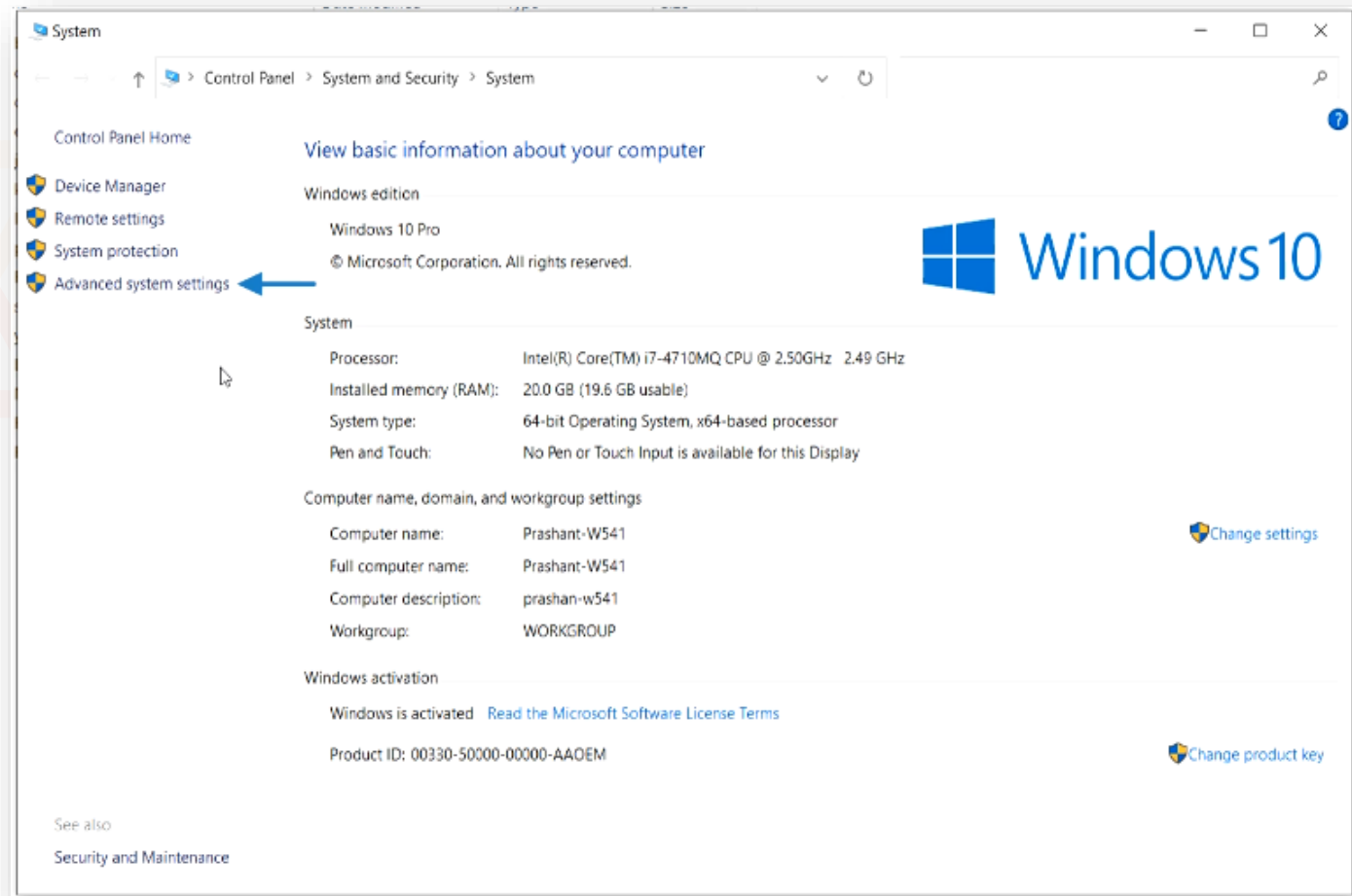
C:\Users\prash>setx PATH "%PATH%;%SPARK_HOME%\bin"

WARNING: The data being saved is truncated to 1024 characters.

SUCCESS: Specified value was saved.

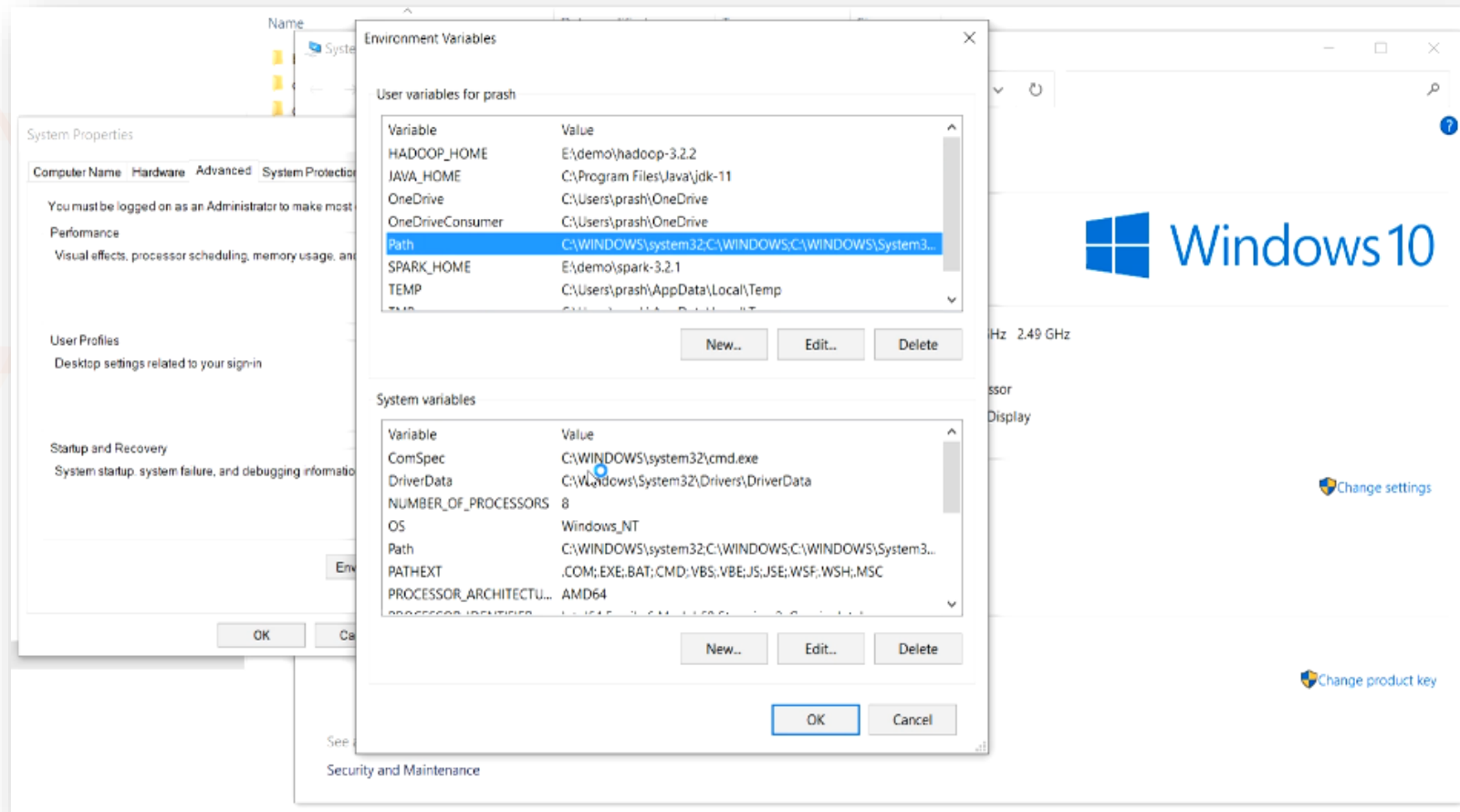
C:\Users\prash>
```

You can add a more extended PATH environment variable using the Windows UI. Right-click on This PC and choose Properties. You will see a window as shown in the image below. Then, go to advanced system settings and then follow the Environment Variable button.

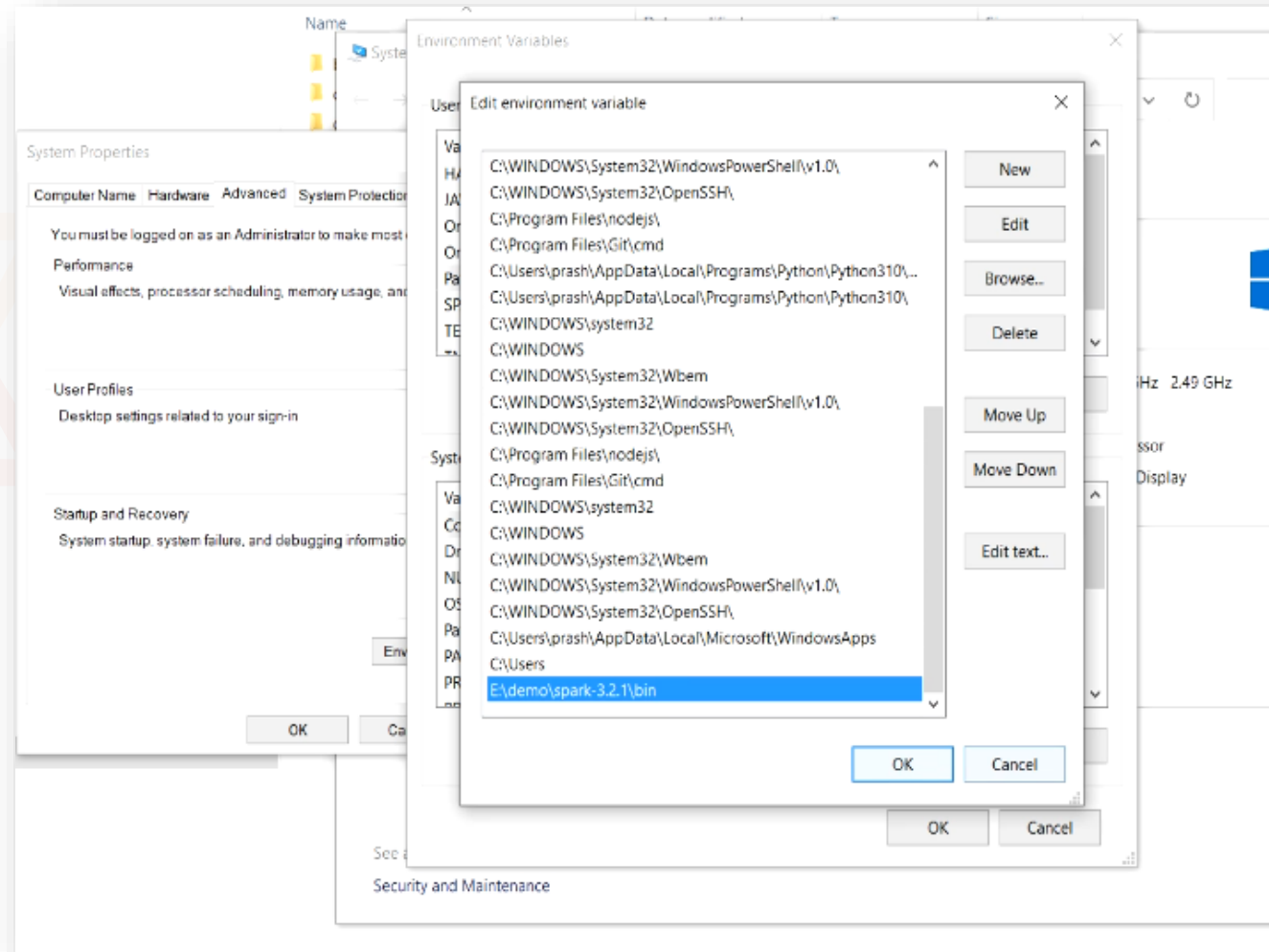


You will land on the window shown below. You will see the Path environment variable.

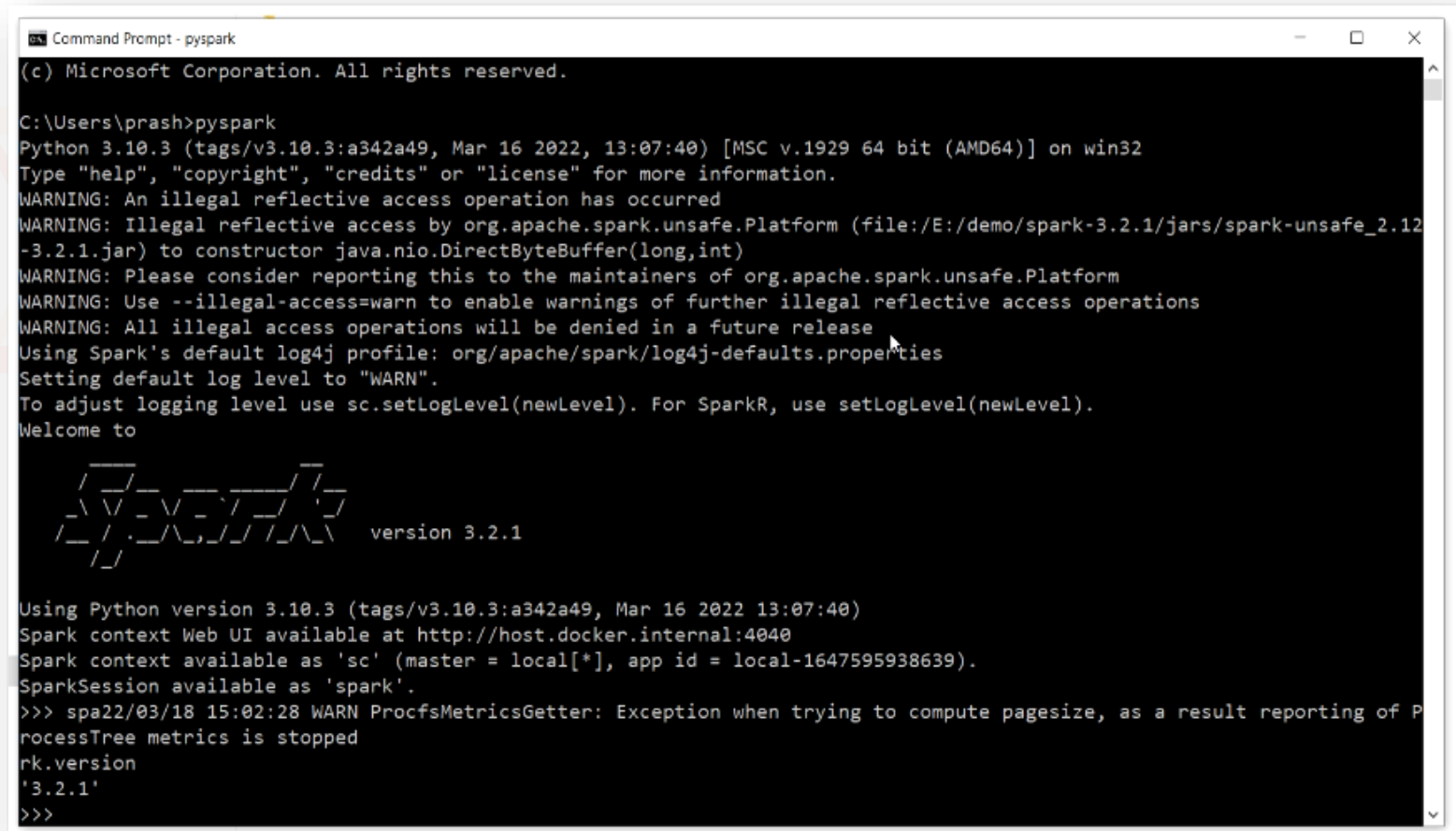
Edit it. You can see a long list of things included in the path ENVIRONMENT Variable.



Click the new button, and paste the Spark Home directory location including the “\bin” at the end. That's all. Click Ok and close everything.



To test if Spark setup is successful, go to command prompt and type “pyspark”
You should see some messages and finally a command prompt. If you see a command prompt and no error messages, you are good to go. You have Apache Spark running on your local machine. You might see a warning stating Exception when trying to compute the page size.



```
(c) Microsoft Corporation. All rights reserved.

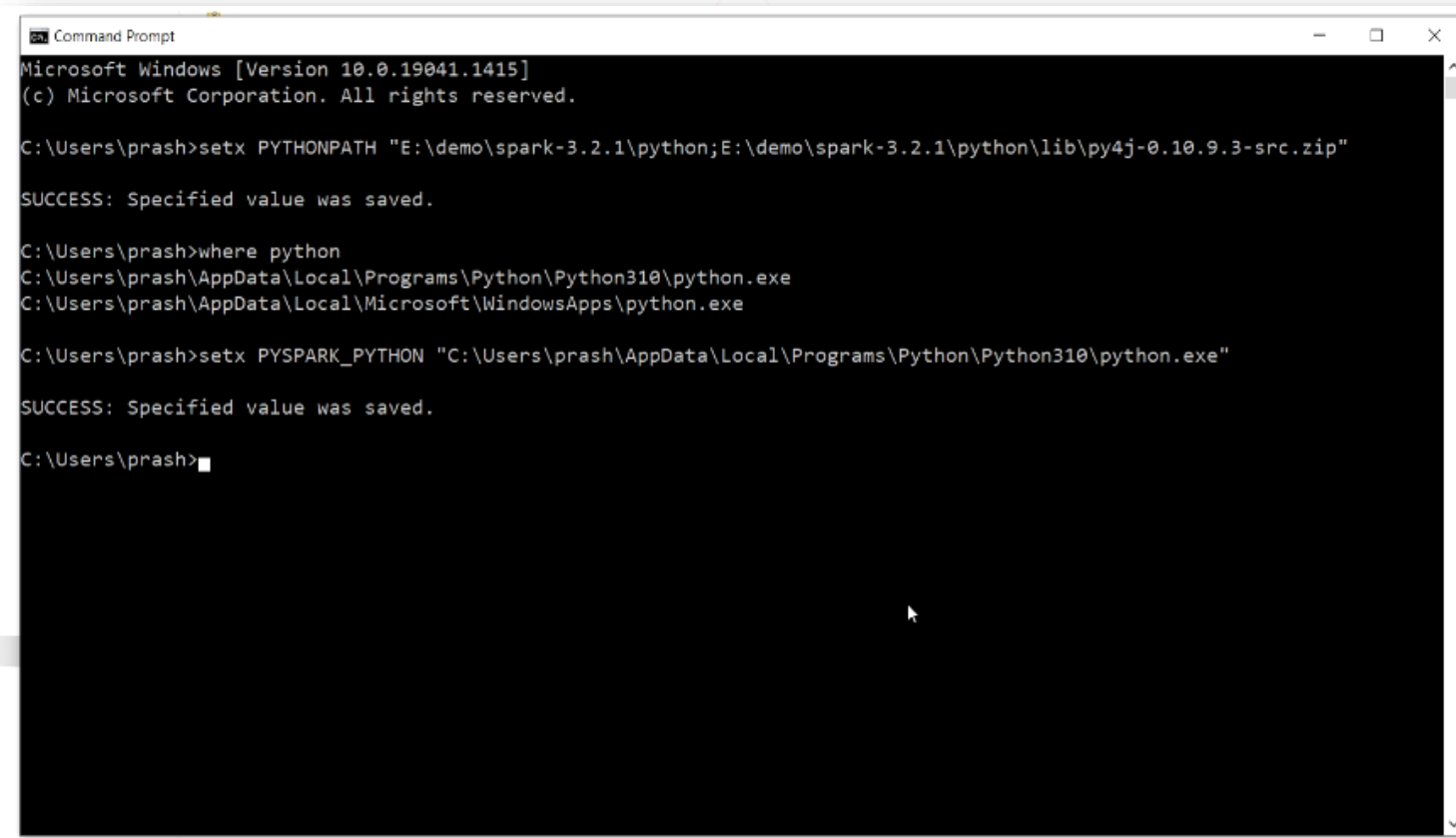
C:\Users\prash>pyspark
Python 3.10.3 (tags/v3.10.3:a342a49, Mar 16 2022, 13:07:40) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.spark.unsafe.Platform (file:/E:/demo/spark-3.2.1/jars/spark-unsafe_2.12-3.2.1.jar) to constructor java.nio.DirectByteBuffer(long,int)
WARNING: Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

  ____      __
 / ___ |__ /  /  ___
/  _ \|_ \|  /  / _ \
/  __/___ \| /  /  __/
/___/_____/ /___/___/
version 3.2.1

Using Python version 3.10.3 (tags/v3.10.3:a342a49, Mar 16 2022 13:07:40)
Spark context Web UI available at http://host.docker.internal:4040
Spark context available as 'sc' (master = local[*], app id = local-1647595938639).
SparkSession available as 'spark'.
>>> spa22/03/18 15:02:28 WARN ProcfsMetricsGetter: Exception when trying to compute pagesize, as a result reporting of ProcessTree metrics is stopped
rk.version
'3.2.1'
>>>
```

The next step is to set up the following two environment variables that can save you from seeing unnecessary errors while working with Spark on your local machine.

1. PYTHONPATH – If you go back to your Spark home directory, you will see a Python directory there. Go inside that python directory and copy that path. Then, Paste it into the PYTHONPATH variable value. Make sure to take the absolute path to the py4j zip file and paste it to PYTHONPATH.
2. PYSPARK_PYTHON - You can use the where-command to find your Python installation location. And use the location to set PYSPARK_PYTHON variable.



```
Command Prompt
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prash>setx PYTHONPATH "E:\demo\spark-3.2.1\python;E:\demo\spark-3.2.1\python\lib\py4j-0.10.9.3-src.zip"

SUCCESS: Specified value was saved.

C:\Users\prash>where python
C:\Users\prash\AppData\Local\Programs\Python\Python310\python.exe
C:\Users\prash\AppData\Local\Microsoft\WindowsApps\python.exe

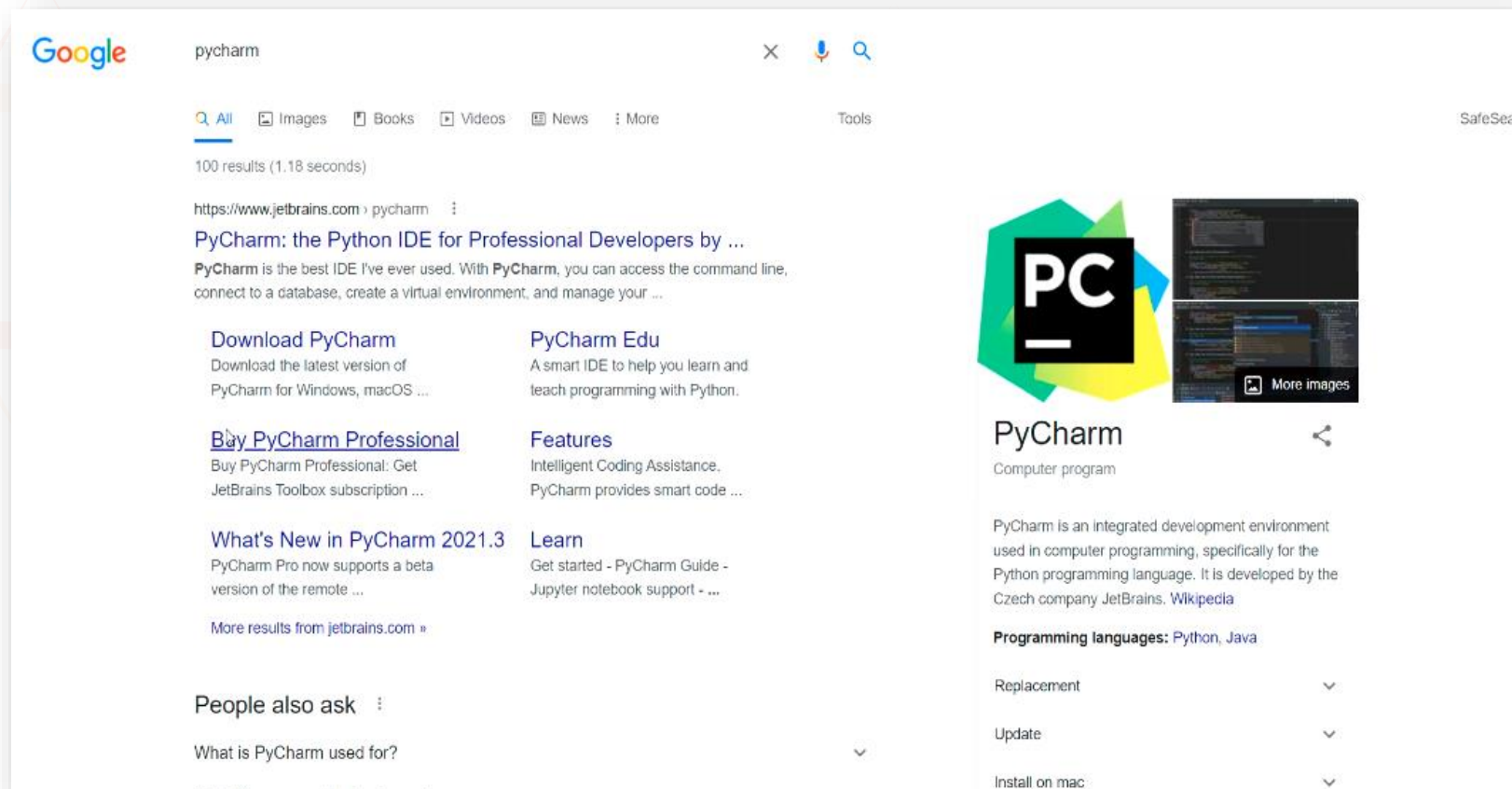
C:\Users\prash>setx PYSPARK_PYTHON "C:\Users\prash\AppData\Local\Programs\Python\Python310\python.exe"

SUCCESS: Specified value was saved.

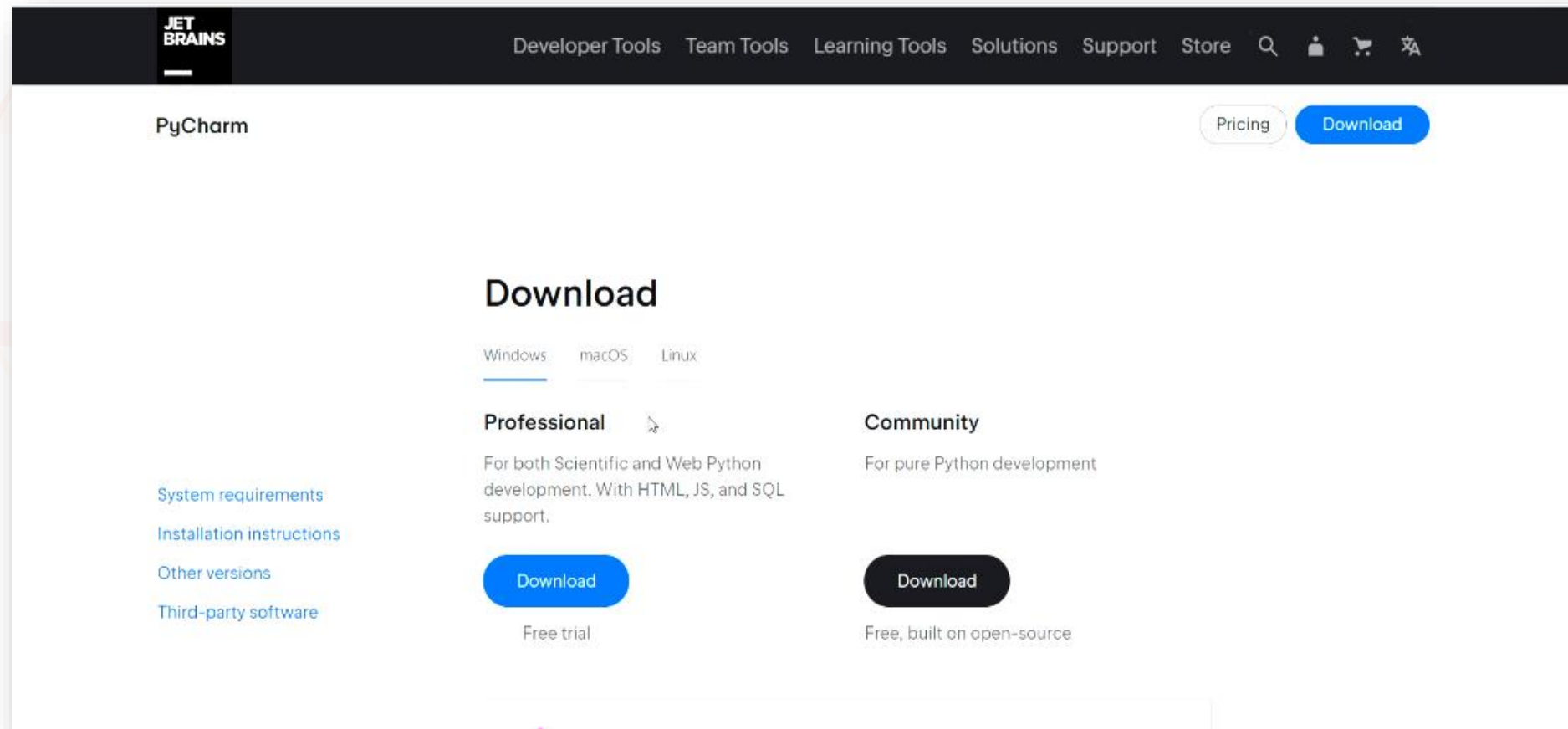
C:\Users\prash>
```


But writing code on command prompt is not a very good approach. We do not want to develop large Spark applications working on the command prompt. I need an IDE for being productive and managing large project development. PyCharm is the most popular IDE for Spark development.

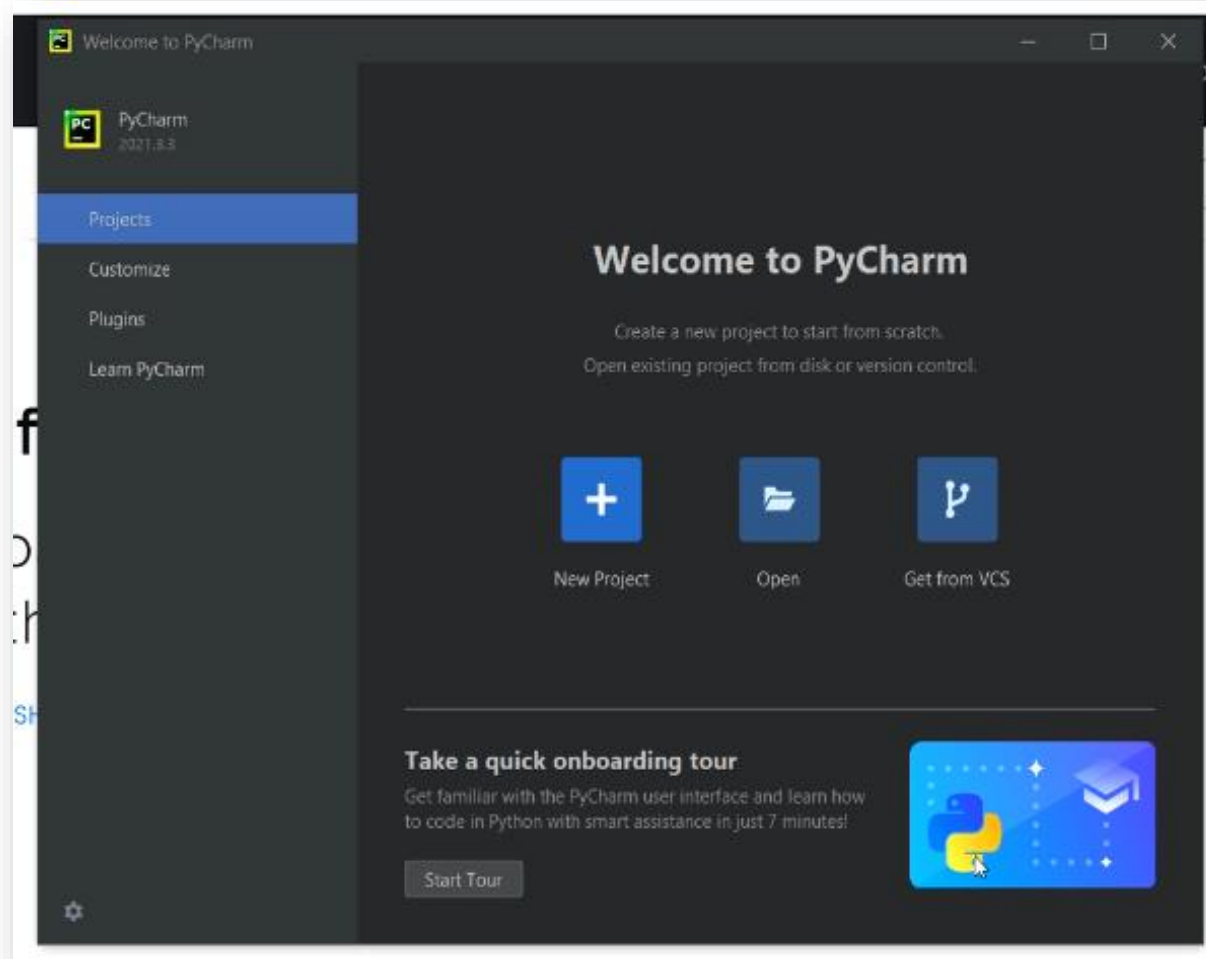
Start your browser and search for PyCharm. And click the link shown jetbrains.com.



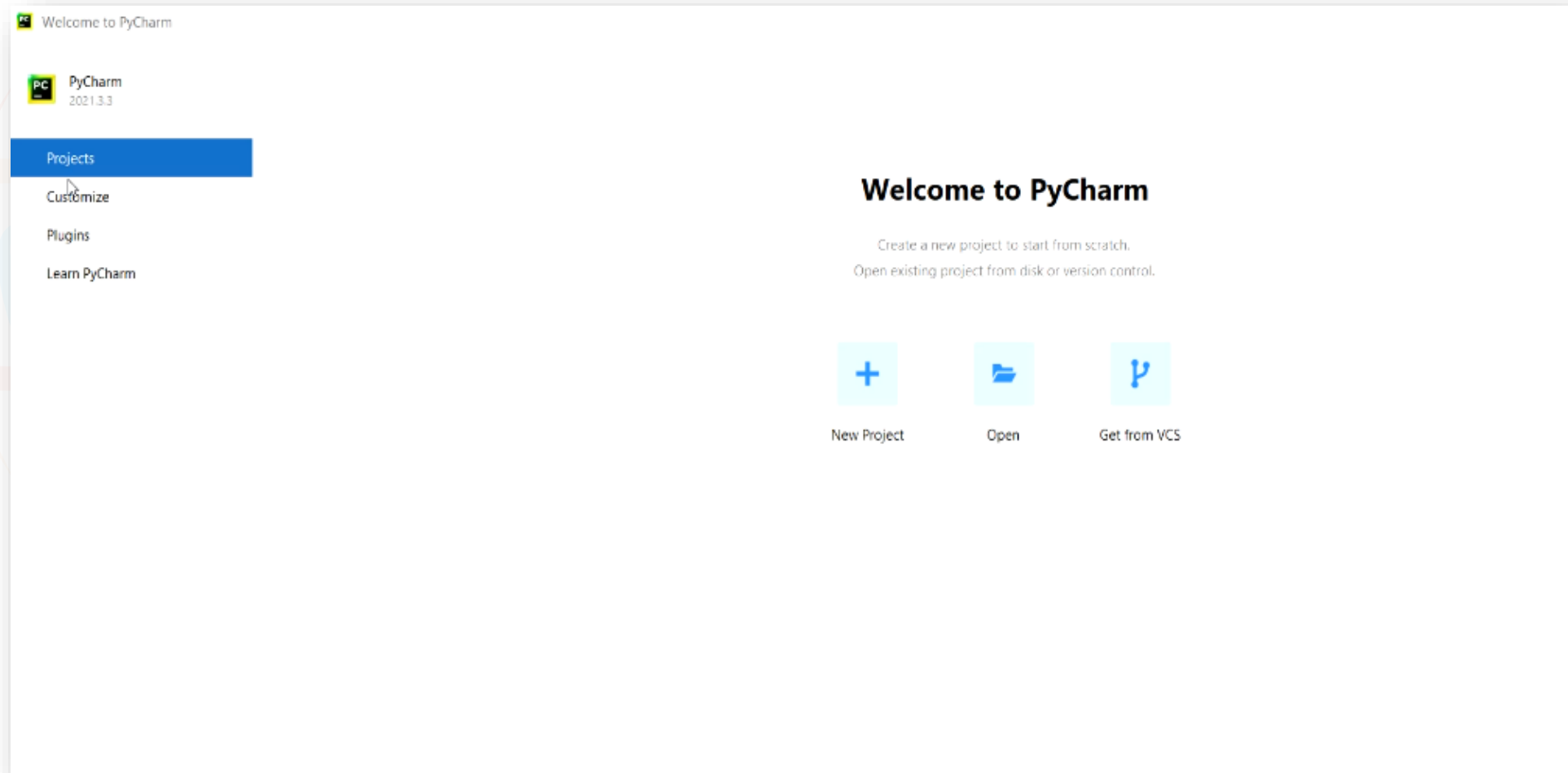
Click on the downloads button, and select the community edition.
The community edition is a free opensource version of PyCharm.
And the community edition is more than enough for our purpose.



Execute the downloaded installer and follow the on-screen default instructions. Once installed, you can start your PyCharm IDE. And you will see the welcome screen as shown below. You can go to customize option to set a lighter theme.



You can come to the Project menu, and you are ready to create your first Spark project.





Thank You
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